

Functional and Non-Functional Requirements in Software Engineering

In software engineering, requirements are crucial for guiding the development process and ensuring that the final product meets user needs. These requirements are typically categorized into **functional** and **non-functional** requirements, each serving distinct purposes.

Functional Requirements

Functional requirements specify what a system should do. They define the specific behaviors, functions, and features that the software must possess to meet user needs. These requirements are often articulated in a straightforward manner to ensure clarity and understanding among stakeholders.

Key Characteristics:

- **Definition**: Describe specific functions or features of the system (e.g., user authentication, transaction processing).
- **Documentation**: Can be documented in various formats such as use cases, user stories, or models.
- **Examples**:
 - Users must be able to log in and log out.
 - The system should send a confirmation email upon registration.
 - Users should be able to search for products using multiple filters.

Functional requirements are essential for verifying that the application fulfils its intended purpose and meets the contractual obligations defined by stakeholders [1][2][3].

Non-Functional Requirements

Non-functional requirements define the quality attributes of a system. They focus on how the system performs its functions rather than what functions it performs. These requirements are critical for ensuring a positive user experience and overall system reliability.

Key Characteristics:

- **Definition**: Specify criteria that can be used to judge the operation of a system (e.g., performance, security, usability).

- **Documentation**: Often captured as quality attributes rather than specific functionalities.
- **Examples**:
 - The system should process requests within 2 seconds.
 - The application must support at least 10,000 concurrent users without performance degradation.
 - Data must be encrypted during transmission.

Non-functional requirements are not mandatory like functional ones but are crucial for achieving a well-rounded product that satisfies user expectations and adheres to industry standards [1][2][3].

Comparison of Functional and Non-Functional Requirements

Aspect	Functional Requirements	Non-Functional Requirements
Definition	What the system should do	How the system performs its functions
Specification	Stated by end-users	Specified by technical teams (architects, developers)
Documentation	Captured in use cases or user stories	Documented as quality attributes
Mandatory	Yes	No
Testing	Functional testing (e.g., unit tests)	Non-functional testing (e.g., performance testing)
Examples	User login functionality	System response time under load

Understanding both functional and non-functional requirements is essential for successful software development. They guide the design process, influence architecture decisions, and ultimately determine whether the software meets its intended goals